Classification of Electrical Faults ⚡⚠

- Based on Symmetry
  - Symmetrical Faults
    - Three-Phase Faults
    - Three-Phase-to-Ground Faults
  - Unsymmetrical Faults
    - Single Line-to-Ground (L–G) Faults
    - Line-to-Line (L–L) Faults
    - Double Line-to-Ground (L–L–G) Faults

  Faults categorized by phase imbalance.

- Based on Duration
  - Transient Faults
    - Temporary in nature
    - Clear automatically (e.g., lightning)
    - Protection schemes may trip and reclose
  - Permanent Faults
    - Persist until isolated or repaired
    - Caused by equipment failure/broken conductors

  Faults categorized by their persistence.

- Based on Nature of Connection
  - Grounded Faults
    - Involve ground contact
    - Examples: L–G or L–L–G faults
  - Ungrounded Faults
    - Occur between conductors without ground
    - Example: L–L faults

  Classification based on contact with system ground.

- Based on Equipment or Location
  - Generator Faults
    - Internal short circuits
    - Stator winding faults
  - Transformer Faults
    - Winding-to-winding or winding-to-ground
    - Core insulation failures
  - Transmission Line Faults — Caused by lightning or pollution
  - Busbar Faults — Insulation failure at substations

  Faults localized by the physical component affected.

- Based on System Response
  - Active Faults
    - Associated with conducting path
    - Involves fault current flow (e.g., short circuits)
  - Passive Faults
    - Do not involve direct current flow
    - Alter system parameters (e.g., open circuits)

  Distinction based on fault current flow.

- Based on Cause of Initiation
  - Natural Faults — Lightning strikes or storms
  - Human-Induced Faults — Operational errors or maintenance oversights
  - Equipment Faults — Aging or insulation deterioration

  Faults categorized by their initiating event source.